Т6. Случайная величина имеет распределение Парето:

$$p(x) = \begin{cases} \dfrac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$$

а) $L(\theta) = \prod_{i=1}^{n} p(x_i; \theta) = (\theta-1)^n \cdot \dfrac{1}{\prod_{i=1}^{n} x_i^\theta} \{(x_i \geq 1)\}$

$\ln L(\theta) = \ln(\theta-1)^n + \ln\left(\prod_{i=1}^{n} x_i\right)^{-\theta} =$

$= n\ln(\theta-1) - \theta\ln\left(\prod_{i=1}^{n} x_i\right)$; $\quad (\ln L(\theta))'_\theta =$

$= \dfrac{n}{\theta-1} - \ln\left(\prod_{i=1}^{n} x_i\right) = 0 \quad \theta - 1 = \dfrac{n}{\sum_{i=1}^{n} \ln x_i} \Rightarrow$

$\Rightarrow \theta = 1 + \dfrac{n}{\sum_{i=1}^{n} \ln x_i}$; $\quad (\ln L(\theta))''_\theta = -\dfrac{n}{(\theta-1)^2} < 0$

$\Rightarrow \max \Rightarrow \boxed{\tilde{\theta} = 1 + \dfrac{n}{\sum_{i=1}^{n} \ln x_i}}$

в) $p_{\xi_{(k)}} = n C_{n-1}^{k-1} F^{k-1} (1-F)^{n-k} \cdot p(t)$ (из Т2)

$$F(x) = \int_{-\infty}^{x} p(t)\,dt = \begin{cases} 0, & x \leq 1 \\ \int_1^x \dfrac{\theta-1}{t^\theta}\,dt, & x > 1 \end{cases} =$$

$$= \begin{cases} 0, & x \leq 1 \\ \theta-1 \int_1^x \dfrac{dt}{t^\theta} = (\theta-1)\dfrac{t^{-\theta+1}}{-\theta+1}\Big|_1^x = \theta-1\left(\dfrac{1}{(1-\theta)x^{-1+\theta}} - \dfrac{1}{1-\theta}\right), & x > 1 \end{cases} =$$

$$= \begin{cases} 0, & x \leq 1 \\ -\dfrac{1}{x^{-1+\theta}} + 1 = 1 - \dfrac{1}{x^{-1+\theta}}, & x > 1 \end{cases}$$

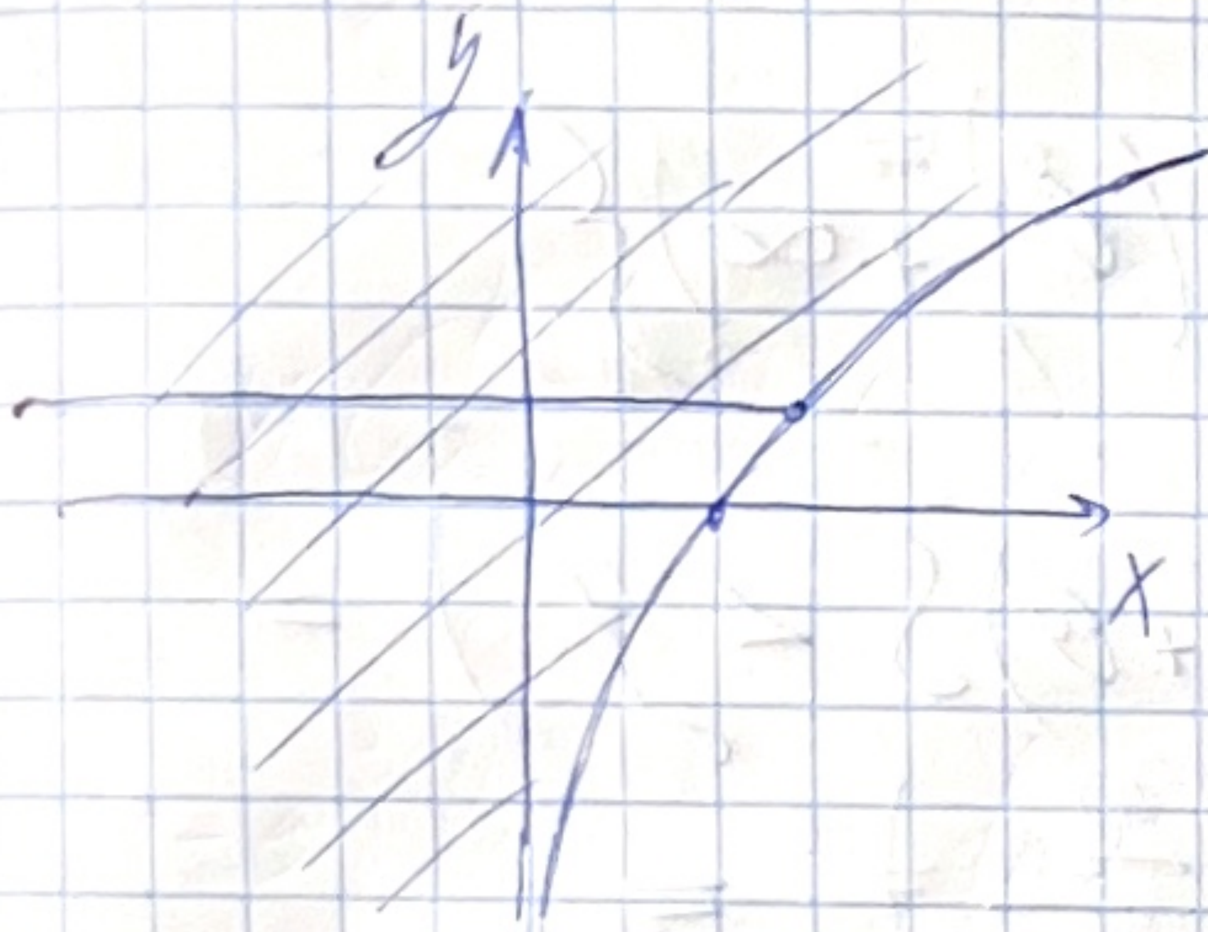
Медиана: $F(x_{1/2}) = 1/2 \Rightarrow 1 - \frac{1}{x_{1/2}^{\theta-1}} = 1/2$

$\Rightarrow x_{1/2}^{\theta-1} = 2 \qquad x_{1/2} = \sqrt[\theta-1]{2}$

$(\theta-1)\ln x_{1/2} = \ln 2$

Рассм. случ. вел. $\eta = \ln\xi$ $\Phi(y) = P(\eta < y) =$

$= P(\ln\xi < y) = \int\limits_{\ln x < y} dF(x) = \int\limits_{-\infty}^{e^y - 0} dF(x) =$



$= F(e^y - 0) - 0 = F(e^y) =$

$= \begin{cases} 0, & e^y \le 1 \\ 1 - \frac{1}{e^{y(\theta-1)}}, & e^y > 1 \end{cases} =$

$= \begin{cases} 0, & y \le 0 \\ 1 - e^{-y(\theta-1)}, & y > 0 \end{cases}; \quad p(y) = \Phi' = \begin{cases} 0, & y \le 0 \\ (\theta-1)e^{-y(\theta-1)}, & y > 0 \end{cases}$

$\Rightarrow \eta_k \sim \Gamma(\theta-1, 1) \quad \left(\frac{(\theta-1)^1}{\Gamma(1)} x^{1-1} e^{-(\theta-1)x} = \right.$

$= (\theta-1)e^{-x(\theta-1)} \{(0; +\infty)\}; \quad \Gamma(1) = \int_0^\infty e^{-x} dx =$

$\left. = -e^{-x}\Big|_0^\infty = 1 \right)$

Покажем, что η_k – независ. Т.к. ξ_k – независ. как э-ты выборки, то

$\Phi(y_1, \dots, y_k) = P(\eta_1 < y_1, \dots, \eta_k < y_k) = P(\ln\xi_1 < y_1, \dots,$
$\ln\xi_k < y_k) = \iint\limits_{\substack{\ln x_1 < y_1 \\ \ln x_k < y_k}} \dots \int dF(x_1, \dots, x_k) = \iint\limits_{\substack{\ln x_1 < y_1 \\ \ln x_k < y_k}} \dots \int p(x_1, \dots, x_k)\, dx =$

$\overset{\text{т.к. независ.}}{=} \iint\limits_{\substack{\ln x_1 < y_1 \\ \ln x_k < y_k}} \dots \int p(x_1) \dots p(x_k)\, dx_1 \dots dx_k \; =$

$$\ominus \left[\begin{array}{l} 0_{e^{y_1}\to 0}, \text{ если } \exists y_i : y_i \underset{e^{y_i}\to 0}{\le 0} \\ \int\limits_{-\infty} p(x_1)dx_1 \int\limits_{-\infty} p(x_2)dx_2 \ldots \int\limits_{-\infty} p(x_k)dx_k = \Phi_{\eta_1}(y_1)\cdot\ldots\cdot\Phi_{\eta_k}(y_k) \end{array}\right. \Rightarrow \text{независ.}$$

$\Rightarrow$ по св-ву суммы: $\zeta = \eta_1 + \ldots + \eta_n = S =$

$$= \sum_{i=1}^{n} \ln \xi_i \sim \Gamma(\theta-1, n)$$

$$p_{\Gamma(\theta-1,n)}(x) = \frac{(\theta-1)^n}{\Gamma(n)} x^{n-1} e^{-(\theta-1)x} \{(0, +\infty)\}$$

Расм. $2(-1+\theta)S$: $\tilde{F} = P(2(-1+\theta)S < y) =$

$$= P\left(S < \frac{y}{2(-1+\theta)}\right) = F\left(\frac{y}{2(-1+\theta)}\right) =$$

$$= \int\limits_{-\infty}^{-\frac{y}{2(1-\theta)}} p_{\Gamma(\theta-1;n)}(x)dx = \int\limits_{0}^{-\frac{y}{2(1-\theta)}} \frac{(\theta-1)^n}{\Gamma(n)} x^{n-1} e^{-(\theta-1)x} dx =$$

$$= \left\{\begin{array}{l} -2(1-\theta)x = t \\ -2(1-\theta)dx = dt \\ x = t/_{-2(1-\theta)} \end{array}\right\} = \int\limits_{0}^{y} \frac{(\theta-1)^n}{\Gamma(n)} \frac{t^{n-1}}{2^{n-1}(\theta-1)^{n-1}} e^{-\frac{(\theta-1)t}{2(\theta-1)}} \frac{dt}{2(\theta-1)} =$$

$$= \int\limits_{0}^{y} \frac{t^{n-1}}{2^n\Gamma(n)} e^{-t/2} dt \Rightarrow \tilde{p}(x) = \tilde{F}' = \frac{x^{n-1}}{2^n\Gamma(n)} e^{-x/2} \{(0; +\infty)\}$$

$$\Rightarrow 2(\theta-1)S \sim \chi^2(2n)$$

Таким образом, $t(\vec{X}_n, x_{1/2}) = 2\frac{\ln 2}{\ln x_{1/2}} S =$

$$= \frac{2\ln 2}{\ln x_{1/2}} \cdot \sum_{i=1}^{n} \ln x_i \sim \chi^2(2n)$$

$$t_1 = \chi^2_{\frac{1-\beta}{2}}(2n), \quad t_2 = \chi^2_{\frac{1+\beta}{2}}(2n)$$

$$P\left(t_1 < \frac{2\ln 2}{\ln x_{1/2}} S < t_2\right) = \beta$$

$$\Rightarrow P\left(\frac{2\ln 2\cdot S}{t_2} < \ln X_{1/2} < \frac{2\ln 2\cdot S}{t_1}\right) = \beta$$

$$\left(e^{\ln 2}\right)^{\frac{2S}{t}}$$

$$\boxed{e^{\frac{2\ln 2\cdot S}{t_2}} < X_{1/2} < e^{\frac{2\ln 2\cdot S}{t_1}}}$$ – доверительный интервал для медианы

$$\Downarrow$$

$$2^{\frac{2S}{t_2}} < X_{1/2} < 2^{\frac{2S}{t_1}}$$

c) $\Theta = (1; +\infty)$ – открытое мн-во

Проверка сильной регулярности модели:

1) $F(x,\theta) = \begin{cases} 0, & x \le 1 \\ 1 - \frac{1}{x^{\theta-1}}, & x > 1 \end{cases}$ – 2-ды непр. дифф. по θ на $\theta > 1$ ⊕

2) Регулярность модели:

$F(x,\theta)$ – непр. дифф. по θ на $\theta > 1$,

$$\int_1^{+\infty} \frac{\partial p}{\partial \theta}\, dx = \int_1^{+\infty} \frac{x^\theta - (\theta-1)x^\theta \ln x}{x^{2\theta}}\, dx =$$

$$= \int_1^{+\infty}\left(\frac{1}{x^\theta} - \frac{(\theta-1)\ln x}{x^\theta}\right)dx = I_1 - I_2 = \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0 \ ⊕$$

$$I_1 = \int_1^{+\infty} \frac{dx}{x^\theta} = \frac{x^{-\theta+1}}{-\theta+1}\Bigg|_1^{+\infty} = \frac{1}{(1-\theta)x^{\theta-1}}\Bigg|_1^{\infty} = \left\{\begin{matrix}\theta>1\\ \theta-1>0\end{matrix}\right\} = \frac{1}{\theta-1}$$

$$I_2 = \int_1^{+\infty} \frac{(\theta-1)\ln x}{x^\theta}\, dx = \int_1^{+\infty} (\theta-1)\ln x \, d\,\frac{x^{1-\theta}}{1-\theta} =$$

$$= (\theta-1)\left[\frac{\ln x}{(1-\theta)x^{\theta-1}}\Bigg|_1^{+\infty} - \int_1^{+\infty} \frac{1}{(1-\theta)x^{\theta-1}}\cdot\frac{1}{x}\, dx\right] =$$

$$= (\theta-1)\left[-\lim_{x\to+\infty}\frac{1/x}{(\theta-1)^2 x^{\theta-2}} - 0 + \frac{1}{\theta-1}\int_1^{+\infty}\frac{1}{x^{\theta}}\,dx\right] =$$

лопиталь

$$= \frac{\theta-1}{\theta-1}\int_1^{+\infty}\frac{1}{x^{\theta}}\,dx = \frac{1}{\theta-1}$$

$$I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial\theta}\right)^2\right] =$$

$$p(x,\theta) = \frac{\theta-1}{x^{\theta}}\{[1;+\infty)\}; \quad \ln p(x,\theta) = \ln(\theta-1) - \theta\ln x$$

$$\frac{\partial \ln p}{\partial\theta} = \frac{1}{\theta-1} - \ln x, \quad \left(\frac{\partial \ln p}{\partial\theta}\right)^2 = \frac{1}{(\theta-1)^2} - \frac{2\ln x}{\theta-1} + (\ln x)^2$$

$$= \int_1^{+\infty}\left(\frac{1}{(\theta-1)^2} - \frac{2\ln x}{\theta-1} + (\ln x)^2\right)\frac{\theta-1}{x^{\theta}}\,dx =$$

$$= \int_1^{+\infty}\frac{1}{(\theta-1)x^{\theta}}\,dx - \int_1^{+\infty}\frac{2\ln x}{x^{\theta}}\,dx + \int_1^{+\infty}\frac{(\theta-1)\ln^2 x}{x^{\theta}}\,dx =$$

$$= I_1 - 2I_2 + (\theta-1)I_3 =$$

$$I_1 = \int_1^{+\infty}\frac{1}{\theta-1\,x^{\theta}}\,dx = \frac{1}{(\theta-1)^2}, \quad I_2 = \int_1^{+\infty}\frac{\ln x}{x^{\theta}}\,dx = \frac{1}{(\theta-1)^2}$$

$$I_3 = \int_1^{+\infty}\frac{\ln^2 x}{x^{\theta}}\,dx = \int_1^{+\infty}\ln^2 x\, d\left(\frac{x^{1-\theta}}{1-\theta}\right) =$$

$$= \frac{x^{1-\theta}}{1-\theta}\ln^2 x\Big|_1^{+\infty} - \int_1^{+\infty}\frac{1}{(1-\theta)x^{\theta-1}}\frac{2\ln x}{x}\,dx =$$

$$= \frac{2}{\theta-1}\int_1^{+\infty}\frac{\ln x}{x^{\theta}}\,dx = \frac{2}{(\theta-1)^3}$$

$$= \frac{1}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2}$$ — непрер. и положит. на $\theta > 1$

$\Rightarrow$ модель регулярна $\oplus$

3) $$\frac{\partial^2}{\partial\theta^2}\int\limits_{-\infty}^{+\infty} dF(x,\theta) \overset{?}{=} \int\limits_{-\infty}^{\infty} d\,\frac{\partial^2 F(x,\theta)}{\partial\theta^2}$$

$$\frac{\partial^2}{\partial\theta^2}\int\limits_{-\infty}^{+\infty} dF(x,\theta) = \frac{\partial^2}{\partial\theta^2}(1) = 0$$

$$\frac{\partial F}{\partial\theta} = \frac{\partial}{\partial\theta}\left(1-\frac{1}{x^{\theta-1}}\right) = \frac{\partial}{\partial\theta}\left(1-x^{1-\theta}\right) =$$

$$= + x^{1-\theta}\ln x \quad , \quad \frac{\partial^2 F}{\partial\theta^2} = -x^{1-\theta}\ln^2 x$$

$$d\left(\frac{\partial^2 F}{\partial\theta^2}\right) = -\left((1-\theta)x^{-\theta}\ln^2 x + x^{1-\theta}\frac{2\ln x}{x}\right)dx =$$

$$= \left(\frac{(\theta-1)\ln^2 x}{x^\theta} - \frac{2\ln x}{x^\theta}\right)dx \quad ; \quad \left\{\frac{\partial^2 F}{\partial\theta^2} = -x^{1-\theta}\ln^2 x\ \{(1;+\infty)\}\right\}$$

$$\int\limits_{-\infty}^{+\infty} = \int\limits_{1}^{+\infty}\left(\frac{(\theta-1)\ln^2 x}{x^\theta} - \frac{2\ln x}{x^\theta}\right)dx = \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0 \ \oplus$$

$\Rightarrow$ Модель явл. сильно регулярной.

$$g(\theta) = \theta \ , \quad \nabla g = 1 \ , \quad I(\theta) = \frac{1}{(\theta-1)^2}$$

$$\frac{\sqrt{n}\,(\hat\theta-\theta)}{\sigma(\hat\theta)} \rightsquigarrow N(0;1) \ , \quad \sigma^2(\theta) =$$

$$= \nabla g^T I^{-1}(\theta)\nabla g = (\theta-1)^2 \Rightarrow \sigma = \theta - 1$$

$$\sqrt{n}\,\frac{\hat\theta-\theta}{\hat\theta-1} \rightsquigarrow N(0,1) \ , \quad \hat\theta = 1 + \frac{n}{\sum \ln x_i}$$

$$t_1 = u_{\frac{1-\beta}{2}} \ , \quad t_2 = u_{\frac{1+\beta}{2}}$$ — квантили

$$P\left(t_1 < \sqrt{n}\,\frac{\hat{\theta} - \theta}{\hat{\theta} - 1} < t_2\right) = \beta$$

$$\frac{t_1(\hat{\theta} - 1)}{\sqrt{n}} < \hat{\theta} - \theta < \frac{t_2(\hat{\theta} - 1)}{\sqrt{n}}$$

$$\Downarrow$$

$$\boxed{\hat{\theta} - \frac{t_2(\hat{\theta} - 1)}{\sqrt{n}} < \theta < \hat{\theta} - \frac{t_1(\hat{\theta} - 1)}{\sqrt{n}}}$$ - асим. довер. интервал

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Сравнение интервалов:
θ = 2
Доверительный интервал для медианы     : 1.784301 < median < 2.357835, длина = 0.573534
Непараметрический бутстрап для медианы : 1.689177 < median < 2.268982, длина = 0.579805
Параметрический бутстрап для медианы   : 1.698626 < median < 2.255045, длина = 0.556420
Асимптотический доверительный интервал для параметра θ : 1.798534 < θ < 2.187861, длина = 0.389326
Непараметрический бутстрап для параметра θ             : 1.743338 < θ < 2.166901, длина = 0.423563
Параметрический бутстрап для параметра θ               : 1.764372 < θ < 2.160350, длина = 0.395978
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